

3D-Printed Coaxial-Fed Patch Antenna-Embedded Substrates for 5G Antenna-In-Package Applications

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Novelty of work

This work demonstrates a fully 3D-printed, monolithically integrated coaxial-fed antenna-embedded lid substrate. The coaxial feeding structure provides immunity to electromagnetic interference, a significant advantage in compact mmWave AiP applications, and achieves 10.08 GHz impedance bandwidth and 4.40 dBi realized gain at 28 GHz through the integrated CPW-coaxial-patch antenna path.

Abstract Text

5G communication systems are rapidly evolving to support data-intensive applications such as artificial intelligence in edge computing and the Internet of Things. These emerging systems require compact, high-efficiency antennas that can be directly integrated with RF or baseband integrated circuits (ICs) to minimize signal loss at millimeter-wave (mmWave) frequencies. Conventional fabrication for antenna-in-package (AiP) relies on subtractive processes—such as lithography-based patterning, epoxy molding, and high-depth etching to connect top and bottom, requiring de-smear steps—resulting in high cost and long production cycles. Recently, three-dimensional (3D) printing, which is additively manufacturable, offers a promising alternative for its cost-efficiency, free-form design availability, and rapid prototyping. 3D printing enables concurrent fabrication of RF components such as transmission line, vertical feeding vias, and patch antenna, even with a cavity structure for IC chips embedding with one printing, so that the fabrication process is significantly simplified without an additional drilling process. However, signal transmission through a conventional via-fed structure, where the signal via is not fully surrounded by GND, can suffer from significant electromagnetic interference from nearby components or adjacent feeding lines. Although quasi-coaxial feeding lines offer an effective approach to reduce such interference, their incomplete shielding still allows coupling. Moreover, their relatively large footprint, often requiring at least four GND vias to ensure signal integrity, limits their applicability in compact mmWave packages. In this work, we implemented a 3D-printed coaxial-line-embedded lid substrate seamlessly connecting the patch antenna for 28 GHz AiP packaging (**Figure 1a**). Each RF component is individually designed to match its characteristic impedance, and the integrated structure is evaluated using EM simulation (CST Studio Suite, SIMULIA) to identify and mitigate unexpected mismatches or discontinuities at junctions, maintaining optimal system-level RF characteristics. To investigate the impact of feeding structures, via, quasi-coaxial, and coaxial feeding structures are evaluated for their intrinsic losses, and EM simulations are conducted to assess crosstalk when these structures are implemented in antenna arrays. Compared to the conventional via-fed patch antenna, which achieves an impedance bandwidth of 3.79 GHz and a realized gain of 5.26 dBi at 28 GHz, the coaxial-embedded design provides a broader bandwidth of 5.05 GHz and a slightly improved gain of 5.31 dBi, indicating enhanced performance in the mmWave band. The substrate measures $5.2 \times 5.2 \text{ mm}^2$ with a thickness of 1.1 mm. The signal via is 200 μm in diameter, spaced 320 μm from

the GND, which is 100 μm thick. The center-to-center distance between the signal via and the patch antenna is 600 μm . A square patch antenna measuring $2.6 \times 2.6 \text{ mm}^2$ is designed to resonate at 28 GHz. As the signal radiation and transmission characteristics are easily changed, the physical dimension of the designed patch antenna is slightly modified to $2.52 \times 2.52 \text{ mm}^2$ when assembled with a coplanar waveguide line (CPW) to feed, which has a 50 μm -wide signal line and an 85 μm spacing to GND (**Figure 1b**). The monolithically integrated AiP lid substrate, which incorporates a patch antenna, coaxial feeding line, and CPW line, exhibits a wide impedance bandwidth of 10.08 GHz, ranging from 24.67 to 34.75 GHz. The antenna shows a sharp resonance at -38.76 dB near 27.8 GHz, which is within 1% of the target frequency of 28 GHz. The realized gain—a key figure of merit reflecting the product of directivity, radiation efficiency, and impedance matching efficiency—reaches 4.91 dBi (**Figure 1c**). The AiP lid substrate is fabricated using a projection micro-stereolithography 3D printer (S230A, BMF) with high resolution. Metal layers are sequentially deposited by thermal evaporation and electroless plating, ensuring strong adhesion and uniform coverage on the printed substrate. The fabricated antenna with seamlessly integrated feeding lines for AiP is characterized using a network analyzer for S-parameters, and its radiation pattern is measured in an anechoic chamber. This study demonstrates the potential of 3D-printed fabrication for producing AiP lid substrates integrated with RF components, such as antennas and coaxial lines, to enable compact and efficient solutions for next-generation wireless systems.

Figure

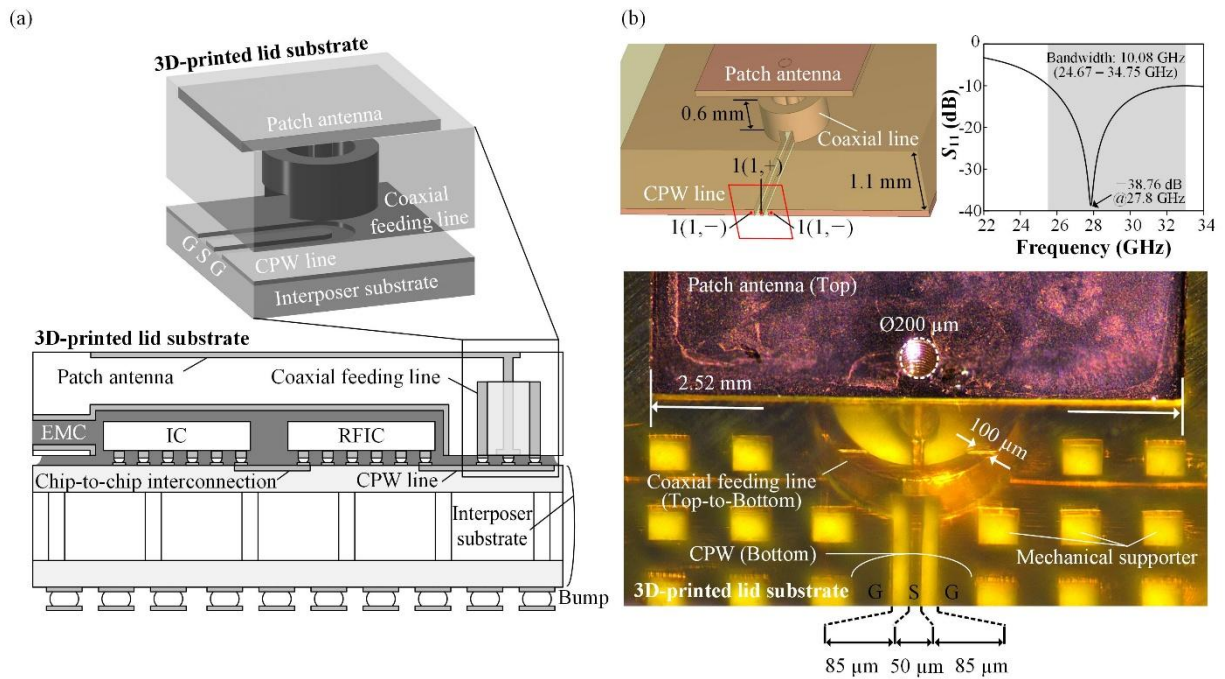


Figure 1. 3D-printed AiP lid substrate with an embedded coaxial-fed patch antenna.

(a) Schematic of the proposed AiP lid substrate with a monolithically integrated coaxial feeding line. (b) 3D modeling, S_{11} simulation result, and image of the fabricated coaxial-fed patch antenna after metallization.